1. Install and set up a DNS server on your local machine or a VM using BIND (Linux) or Simple DNS Plus (Windows), then create a new DNS zone for the domain musicmania.in.

1. Install BIND

Open Terminal:

sudo apt update

sudo apt install bind9 bind9utils bind9-doc -y

Check that BIND is running:

sudo systemctl status bind9

If it is not running:

sudo systemctl start bind9

sudo systemctl enable bind9

2. Create the DNS Zone

Edit the BIND local configuration:

sudo nano /etc/bind/named.conf.local

Add:

```
zone "musicmania.in" {
    type master;
    file "/etc/bind/db.musicmania.in";
};
```

Save with **Ctrl+O** → **Enter**, then **Ctrl+X**.

3. Create the Zone File

Copy the default zone file:

sudo cp /etc/bind/db.empty /etc/bind/db.musicmania.in

Open it:

sudo nano /etc/bind/db.musicmania.in

Replace its contents with:

```
$TTL    604800
@       IN      SOA     ns1.musicmania.in. admin.musicmania.in. (
                        2026081101
                        604800
                        86400
                        2419200
                        604800 )

@       IN      NS      ns1.musicmania.in.
ns1     IN      A       192.168.1.10

@       IN      A       192.168.1.10
www     IN      A       192.168.1.10
mail    IN      A       192.168.1.11
```

Important: Replace 192.168.1.10 with the IP address of your Linux VM/server.

This creates:

Hostname	**IP Address**
musicmania.in	192.168.1.10
www.musicmania.in	192.168.1.10
ns1.musicmania.in	192.168.1.10
mail.musicmania.in	192.168.1.11

4. Check the Configuration

Run:

```
sudo named-checkconf
```

If there is no output, the configuration is correct.

Check the zone:

```
sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in
```

You should see something similar to:

zone musicmania.in/IN: loaded serial 2026081101

OK

5. Restart BIND

sudo systemctl restart bind9

Check again:

sudo systemctl status bind9

6. Test DNS Resolution

Install DNS testing tools if necessary:

sudo apt install dnsutils -y

Test:

dig @127.0.0.1 musicmania.in

Test the www record:

dig @127.0.0.1 www.musicmania.in

You should see an **A record** pointing to:

192.168.1.10

You can also use:

nslookup musicmania.in 127.0.0.1

7. Optional: Configure the Client to Use Your DNS Server

On another Linux machine, set the DNS server to your BIND server's IP:

DNS Server: 192.168.1.10

Then test:

nslookup www.musicmania.in 192.168.1.10

Expected result:

Name: www.musicmania.in

Address: 192.168.1.10

Final Result

Your DNS server now has the zone:

musicmania.in

|

+--- ns1.musicmania.in → 192.168.1.10

|

+--- www.musicmania.in → 192.168.1.10

|

+--- mail.musicmania.in → 192.168.1.11

For a practical/assignment screenshot, capture:

1. systemctl status bind9
2. named-checkzone musicmania.in /etc/bind/db.musicmania.in
3. nslookup www.musicmania.in 127.0.0.1 or dig
4. The /etc/bind/db.musicmania.in zone file.

2. Add an A record in your DNS server configuration to map www.musicmania.in to the IP address 192.168.1.50, then use the nslookup or dig command to verify the mapping works.

1. Edit the DNS zone file

Open the zone file for musicmania.in:

sudo nano /etc/bind/db.musicmania.in

Add this **A record**:

www IN A 192.168.1.50

For example:

$TTL 604800

@ IN SOA ns1.musicmania.in. admin.musicmania.in. (

2

604800

86400

2419200

604800)

@ IN NS ns1.musicmania.in.

ns1 IN A 192.168.1.10

www IN A 192.168.1.50

2. Check the BIND configuration

sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in

You should see something similar to:

zone musicmania.in/IN: loaded serial 2

OK

Also check the overall configuration:

sudo named-checkconf

If there is no output, the configuration is valid.

3. Restart BIND

sudo systemctl restart bind9

Check that it is running:

sudo systemctl status bind9

4. Verify using nslookup

Run:

nslookup www.musicmania.in 127.0.0.1

Expected result:

Server: 127.0.0.1

Address: 127.0.0.1#53

Name: www.musicmania.in

Address: 192.168.1.50

5. Or verify using dig

dig @127.0.0.1 www.musicmania.in

Look for:

;; ANSWER SECTION:

www.musicmania.in.   604800   IN   A   192.168.1.50

Result

The DNS mapping is:

www.musicmania.in → 192.168.1.50

3. Simulate a typo by adding a CNAME record for play.musicmania.in that points to www.musicmania.in, then test if accessing play.musicmania.in resolves to the correct IP address.

1. Edit the zone file

sudo nano /etc/bind/db.musicmania.in

Add:

play   IN   CNAME   www.musicmania.in.

Your records should now look like:

@   IN   NS   ns1.musicmania.in.

ns1   IN   A   192.168.1.10

www   IN   A   192.168.1.50

play   IN   CNAME   www.musicmania.in.

The final . in www.musicmania.in. is important because it specifies the fully qualified domain name.

2. Increase the zone serial number

For example, change:

2026081101

to:

2026081102

3. Check the zone

sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in

Expected:

zone musicmania.in/IN: loaded serial 2026081102

OK

4. Restart BIND

sudo systemctl restart bind9

5. Test the CNAME

Use:

nslookup play.musicmania.in 127.0.0.1

Expected result:

Name: play.musicmania.in

Address: 192.168.1.50

You can also use:

dig @127.0.0.1 play.musicmania.in

Look for:

;; ANSWER SECTION:

play.musicmania.in.   IN   CNAME   www.musicmania.in.

www.musicmania.in.   IN   A   192.168.1.50

Result

play.musicmania.in

↓ CNAME

www.musicmania.in

↓ A record

192.168.1.50

4. Your DNS server is not resolving the domain musicmania.in as expected. Document the step-by-step troubleshooting process you would follow, including checking configuration files, restarting the DNS service, and testing with nslookup or dig.

DNS Troubleshooting Process for musicmania.in

If musicmania.in is not resolving correctly on a **BIND DNS server**, follow these steps in order.

1. Check the DNS service status

sudo systemctl status bind9

If the service is stopped, start it:

sudo systemctl start bind9

Enable it to start automatically:

sudo systemctl enable bind9

2. Check the BIND configuration

Run:

sudo named-checkconf

- **No output** → configuration syntax is correct.
- **Error shown** → check the indicated configuration file and line.

3. Check the DNS zone file

For example:

sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in

Expected output:

zone musicmania.in/IN: loaded serial 2026081102

OK

If you get an error, open the zone file:

sudo nano /etc/bind/db.musicmania.in

Check that records such as these are correctly written:

@ IN NS ns1.musicmania.in.

ns1 IN A 192.168.1.10

www IN A 192.168.1.50

play IN CNAME www.musicmania.in.

Make sure the zone name and nameserver names are correct.

4. Check the BIND zone configuration

Open:

sudo nano /etc/bind/named.conf.local

It should contain something similar to:

```
zone "musicmania.in" {
    type master;
    file "/etc/bind/db.musicmania.in";
};
```

Make sure the file path matches the actual zone file.

5. Check the zone serial number

Inside the zone file, check the SOA record:

```
@ IN SOA ns1.musicmania.in. admin.musicmania.in. (
        2026081102
        604800
        86400
        2419200
        604800
)
```

Whenever you make changes to the zone, **increase the serial number**.

6. Restart the DNS service

After fixing any configuration problems:

sudo systemctl restart bind9

Then check:

sudo systemctl status bind9

The service should show:

Active: active (running)

7. Check DNS port 53

BIND normally listens on port **53**.

Run:

sudo ss -lntup | grep :53

You should see BIND listening on port 53.

You can also check:

sudo ss -lunp | grep :53

This checks UDP port 53, which is commonly used for DNS queries.

8. Test using nslookup

Test the domain:

nslookup musicmania.in 127.0.0.1

Test the www record:

nslookup www.musicmania.in 127.0.0.1

Expected result for www:

Name: www.musicmania.in

Address: 192.168.1.50

Test the CNAME:

nslookup play.musicmania.in 127.0.0.1

It should ultimately resolve to:

192.168.1.50

9. Test using dig

Run:

dig @127.0.0.1 musicmania.in

For the www record:

dig @127.0.0.1 www.musicmania.in

Look for:

;; ANSWER SECTION:

www.musicmania.in.   IN   A   192.168.1.50

For the CNAME:

dig @127.0.0.1 play.musicmania.in

Expected:

;; ANSWER SECTION:

play.musicmania.in.   IN   CNAME   www.musicmania.in.

www.musicmania.in.   IN   A   192.168.1.50

10. Check DNS logs if it still fails

Use:

sudo journalctl -u bind9 --no-pager -n 50

Look for errors such as:

- Zone file syntax error
- Missing zone file
- Permission denied
- Incorrect nameserver
- Failed zone loading

You can also check:

sudo tail -f /var/log/syslog

11. Check firewall

Make sure DNS traffic is allowed on **TCP and UDP port 53**.

For UFW:

sudo ufw status

If necessary:

sudo ufw allow 53/tcp

sudo ufw allow 53/udp

Troubleshooting flow

DNS not resolving

↓

Check BIND service

↓

named-checkconf

↓

named-checkzone

↓

Check named.conf.local

↓

Check zone file & serial number

↓

Restart bind9

↓

Check port 53

↓

nslookup / dig

↓

Check BIND logs

↓

Check firewall

↓

DNS resolution working

Final verification

The final test should show:

www.musicmania.in → 192.168.1.50

play.musicmania.in → www.musicmania.in → 192.168.1.50